Educational Psychology Course as a Learning Opportunity for Improving Pre-service Teachers' Understandings about the Brain

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INTRODUCTION

There is a growing interest in applying neuroscience findings to educational theory, practice, and policy (Dubinsky, Roehrig, & Varma, 2013). However, educators have few opportunities to learn information about the brain. In the absence of *neuroscience literacy*, educators' enthusiasm for neuroscience findings has resulted in the rise of widespread *neuromyths*.

This study evaluated the effect of an introductory educational psychology (EP) course for enhancing neuroscience literacy and reducing beliefs in neuromyths of pre-service teachers. The research questions were:

1. Is neuroscience literacy enhanced when pre-service teachers take an introductory EP course, and are these changes different for US and Korean pre-service teachers?
2. Is belief in neuromyths reduced when pre-service teachers take an introductory EP course, and are these changes different for US and Korean pre-service teachers?
3. What *a priori* factors – academic history and information sources for learning about neuroscience research – impact neuroscience literacy and beliefs in neuromyths in US and Korean pre-service teachers?

METHOD

Participants

- Elementary pre-service teachers in an introductory EP course
- US = 70 (male =7), mostly juniors at UMN (age = 21.47 y)
- Korea = 50 (male =15), all sophomores at SNUE (age = 20. 86 y)
- Participants completed the survey questionnaire twice, at the beginning and end of semester.

Measures

A survey questionnaire was developed largely from items used in prior studies (Dekker et al., 2012; Herculano-Houzel, 2002). It consisted of 60 statements about brain function and its relation to educational practice, 28 of which were correct and 32 incorrect (Cronbach's alpha > .71).

Participants judged each statement as correct ("Yes"), incorrect ("No"), or "I don't know". We also collected background information such as age, gender, academic history (i.e., number of biology and neuroscience courses taken).

Neuroscience Literacy (NL)

- **Neuroscience concepts (NC):** know the basic terminology and functions of the brain
- **Neuroscience processes (NP):** understand how neuroscientists study the brain and produce neuroscience data
- **Neuroscience situations (NS):** apply neuroscience findings to optimize brain function and development to address authentic real-word problems

Neuromyths (NM)

- Misconceptions generated by misinterpretations to make a case for use of brain research in education and other contexts.

RESULTS

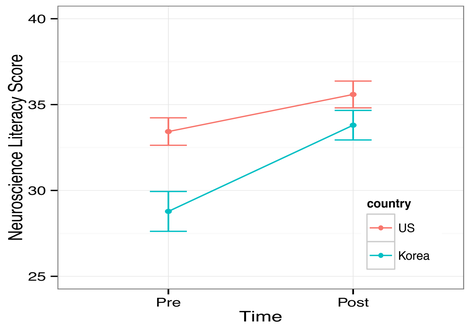


Figure 1. NL scores in the US and Korea

1. Two-way repeated ANOVA found a main effect of Time ($F(1,118)$ = 45.223, $p < .001$, $\eta_p^2 = .277$), a main effect of Country ($F(1,118)$ = 7.813, $p = .006$, $\eta_p^2 = .062$), and a Time-by-Country interaction ($F(1,118)$ = 7.159, $p = .008$, $\eta_p^2 = .057$) (see Figure 1). Separate two-way ANOVAs for the six section scores were represented below (see Table 1).

Table 1. *Effect of taking an introductory EP course on the NL of pre-service teachers*

Section (number of items)	US		Korea		ME Time	ME Country	Time X Country
	Pre *M* (*SD*)	Post *M* (*SD*)	Pre *M* (*SD*)	Post *M* (*SD*)	*p* (η_p^2)	*p* (η_p^2)	*p* (η_p^2)
I. General knowledge (14)	8.20 (2.2)	8.80 (1.8)	7.38 (2.4)	8.02 (2.2)	.003 (.074)	.021 (.044)	*ns*
II. Localization of brain function (8)	4.56 (1.4)	4.27 (1.4)	3.54 (1.4)	4.14 (1.1)	*ns*	.008 (.058)	.001 (.091)
III. Brain development (10)	6.01 (1.5)	6.73 (1.5)	4.42 (2.2)	6.00 (1.6)	< .001 (.273)	< .001 (.143)	.013 (.051)
IV. Brain structure (12)	5.84 (2.1)	6.10 (2.0)	5.42 (2.0)	6.04 (1.8)	.014 (.051)	*ns*	*ns*
V. Neuroimaging (6)	2.56 (1.6)	3.07 (1.7)	2.48 (1.6)	3.46 (1.4)	< .001 (.150)	*ns*	*ns*
VI. Applying neuroscience results (10)	6.26 (1.4)	6.61 (1.6)	5.58 (1.6)	6.14 (1.8)	.013 (.051)	.014 (.050)	*ns*
Total (60)	33.43 (6.7)	35.59 (6.4)	28.78 (8.2)	33.80 (6.0)	< .001 (.277)	.006 (.062)	.008 (.057)

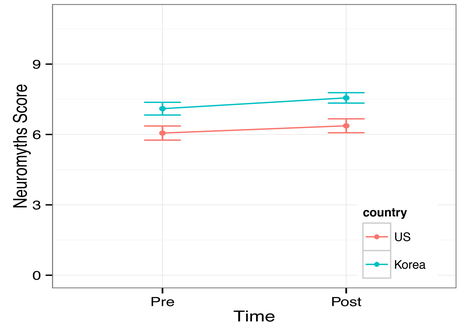


Figure 2. NM scores in the US and Korea

2. Eleven neuromyth items were believed by over 50 % of US or Korean pre-service teachers at pre-test (see Table 2). Two-way repeated ANOVA only found a main effect of country $F(1,118)$ = 13.078, $p < .001$, $\eta_p^2 = .100$). There was no main effect of time ($p = .104$) and Time-by-Country interaction ($p = .758$) (see Figure 2).

Table 2. *NM believed by 50% or more of pre-service teachers at pre- and post- test*

Neuromyth	US		Korea	
	Pre (%)	Post (%)	Pre (%)	Post (%)
I-4. *Blind people have better hearing.*	61	57	80	78
I-13. *Memory is stored in the brain much like as in a computer. That is, each memory is encoded in a tiny piece of the brain.*	61	67	36	54
II-4. *Right-hemisphere learners are more creative than left-hemisphere learners.*	40	40	64	62
II-6. *Brief co-ordination exercises can improve integration of left and right hemispheric brain function.*	63	67	70	86
IV-3. *The volume of blood in the brain increases with physical effort.*	54	47	74	86
….				
IV-10. *Emotional brain processes interrupt cognitive brain processes such as thinking and reasoning.*	61	70	26	24

Note: The numbers of percentage indicate the proportion of incorrect "Yes" responses to each NM.

3-A. Independent samples *t*-tests confirmed that there were no significant differences in neuroscience literacy as a function of whether US pre-service teachers had or had not taken a biology course, whether Korean pre-service teachers had or had not taken a biology course, or whether US pre-service teachers had or had not taken a neuroscience course (see Figure 3).

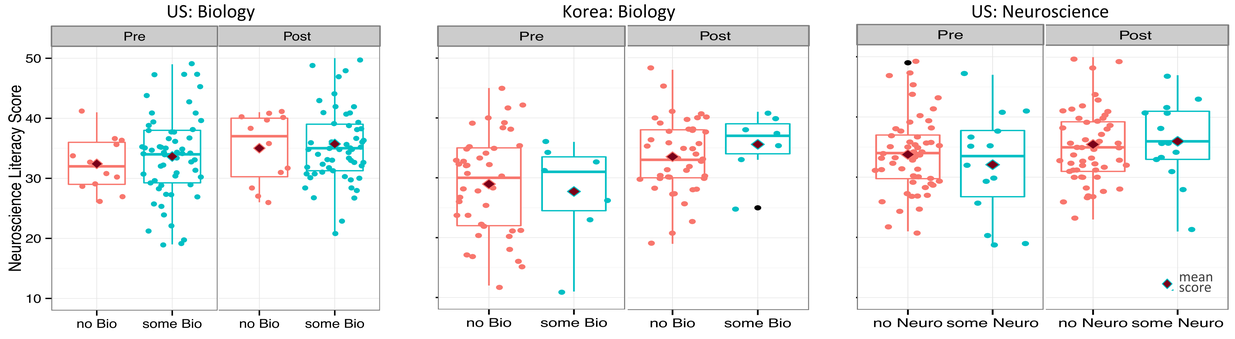


Figure 3. Boxplots showing effects of prior academic history on pre-service teachers' NL

3-B. The main information sources consulted by both groups were books and public media. The proportion of information sources differed by country, χ^2 (5, N = 203) = 18.597, $p = .002$ (see Table 3).

Table 3. *Main information sources about neuroscience in the US and Korea*

	US		Korea		*z*	*p*
	N	%	*N*	%		
Books	55	39.01	13	20.97	2.508	.012
Public Media	29	20.57	25	40.32	-2.934	.003
Scientific magazine and journals	19	13.48	8	12.90	.111	*ns*
Pre-service training	15	10.64	1	1.61	2.198	.028
Commercial products	13	9.22	5	8.06	.267	*ns*
Others	10	7.09	10	16.13	-1.990	.047

Note: *Each participant could indicate multiple information sources.*

DISCUSSION

This study evaluated whether an instructional intervention – taking an introductory EP course – increases NL and decreases beliefs in NM. With respect to our three research questions, we found:

1. The introductory EP course increased NL in five of six domains, with the largest improvement in knowledge of brain development. The course directly target the three components of NL.
   - NC: providing basic knowledge about cognitive development and the brain.
   - NP: introducing on interpretation of student performance.
   - NS: illuminating how psychological theories improve learning and instruction.
2. Taking an introductory EP course did not reduce beliefs in NM. However, a closer examination of the data revealed three classes of NM depending on their persistency. An introductory EP course was not a powerful enough interventions to dispel all NM.
   - ***Discredited* NM**: believed in the past but no longer (e.g. *Mozart effect*)
   - ***Persistent* NM**: believed in the past and continue to be believed today (e.g. *Brain Gym*)
   - ***Emerging* NM**: not measured in prior studies but found to be widely believed (e.g. *Blind people have better hearing*)
3. With respect to moderating factors, information sources from which participants learned about neuroscience findings had a marginal effect of NL, whereas prior coursework had not effect on NL and beliefs in NM. These findings imply that merely knowing the relevant science is not enough to translate neuroscience findings into educational practice.

These findings suggest that an introductory educational psychology course provides another interdisciplinary ground for bridging educators and neuroscientists by providing a learning opportunity for improving pre-service teachers' understandings about the brain.

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